

Hyperspectral Image Compression With Spectral-Spatial Coupling and Group-Wise Context Modeling

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Abstract—The rich spectral information within hyperspectral images (HSIs) results in large data volumes. Thus finding a compact representation for HSIs while maintaining reconstruction quality is a fundamental task for numerous applications. Though the existing learning-based compression methods and context models have shown strong rate-distortion (RD) performance, these methods only pay their attention on spatial redundancy without considering the spectral redundancy of HSIs, which thus impedes further improvement of their performance on HSI. Moreover, the strictly sequential autoregressive nature of context models leads to inefficiency, further limiting their practical applications. In this paper, leveraging the spectral priors unique to HSIs, we propose a hybrid Transformer-CNN architecture to find compact latent representations of HSIs. In specific, we construct Spectral-Spatial Coupling Transformer Group (SSCTG) to cooperatively extract spatial and spectral features of HSIs. Additionally, we propose Group-wise Context Model (GCM) to further enhance the parallel processing capability of autoregression within context models, significantly improving the coding efficiency. Extensive experiments demonstrate the effectiveness of the proposed method, achieving superior RD performance compared to state-of-the-art methods while maintaining high efficiency of codecs.

Index Terms—Hyperspectral image compression, transformer-CNN hybrid architecture, group-wise context model.

I. INTRODUCTION

HYPERSPECTRAL images (HSIs) contain richer spectral information compared to traditional RGB images, thereby enhancing discriminative capabilities. As a result, HSIs have been widely utilized to a wide range of applications such as scene classification [1], target detection [2], environmental monitoring [3], and resource exploration [4]. However, the abundant spectral information increases the data volume substantially, imposing a heavy burden on image

storage and transmission. Thus, effectively compressing HSIs to reduce data redundancy while minimizing data distortion during reconstruction is a challenging and vital research task nowadays.

Traditional HSI compression methods are mainly based on predictive coding [5], [6], [7], vector quantization [8], [9], [10], or transform coding [11], [12] to eliminate spatial and spectral correlations within HSIs. These methods typically consist of multiple independent modules, which operate without interaction and therefore often lead to sub-optimal results. With the development of deep learning, learning-based methods [13], [14], [15], [16], [17], [18], [19], [20] have demonstrated better rate-distortion (RD) performance in image compression. Different from traditional HSI compression methods, learning-based methods jointly optimize all modules in an end-to-end manner, thus demonstrating greater potential over traditional compression methods. Furthermore, [21], [22], [23], [24], [25] have integrated context models into learning-based compression methods and leveraged autoregressive entropy models, further enhancing image compression performance.

The aforementioned compression models typically employ CNN-based architectures as their codecs. With the powerful performance demonstrated by Transformer [26] in the field of computer vision, several studies [27], [28], [29], [30], [31] have pointed out the limitations of the CNN-based compression methods and attempted to introduce Transformer structures into image compression tasks to further improve RD performance. Although existing learning-based compression methods have achieved impressive compression performance, the following two problems still limit further improvement of their performance on HSIs. 1). Compared to RGB images, HSIs contain richer spectral information, leading to a large amount of spectral redundancy. However, existing CNN-based or Transformer-based architectures mainly focused on spatial redundancy in images, neglecting the high similarity and correlation in the spectral dimension. 2). Context models have been demonstrated [16] that can effectively eliminate contextual redundancy but will lead to strict serial processing of encoding and decoding, significantly increasing runtime. Due to the multitude of bands in HSIs ranging from dozens to hundreds, this issue will be further exacerbated, hindering further applications of HSI compression.

In this paper, to address the above two issues of existing compression methods applied to HSIs, we propose a

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learning-based compression method tailored for HSIs. We first delve into exploring effective ways of utilizing Transformer architectures to capture spectral redundancy in HSIs. We design a hybrid Transformer-CNN architecture termed the Spectral-Spatial Coupling Transformer Group (SSCTG), which serves as the core codec structure of the proposed compression method. SSCTG utilizes spectral-wise multi-head self-attention to effectively explore the spectral features of HSIs. Meanwhile, it employs a CNN-based feed-forward network to capture spatial local similarity, enabling the cooperative extraction of both spatial and spectral features of HSIs. Meanwhile, since the strict serial processing of the autoregressive entropy model leads to inefficient decoding, we further propose the Group-wise Context Model (GCM) to reduce the time required for entropy model. We group the latent representations along the channel dimension and only employ autoregressive context models for prediction within each channel group, while processing between channel groups remains parallel, greatly reducing the decoding time complexity. In summary, the main contributions of our work are as follows:

- 1) We propose a learning-based compression method crafted specifically for HSIs with the Spectral-Spatial Coupling Transformer Group (SSCTG) to fully capture both spatial and spectral redundancy of HSIs.
- 2) We propose a Group-wise Context Model (GCM) which enables parallel encoding and decoding across groups to enhance the efficiency of autoregressive context models for HSI compression.
- 3) Extensive experiment results demonstrate that our proposed method outperforms state-of-the-art approaches in terms of RD performance and exhibits higher encoding and decoding efficiency.

II. RELATED WORKS

A. Learning-Based Compression

Recently, learning-based compression methods have achieved promising performance on RGB images. Ballé et al. [13] first proposed an end-to-end learned model and outperformed JPEG2000 [12] in RD performance, initiating the trend of learning-based image compression. Based on variational auto-encoder (VAE) [32], Ballé et al. [33] further incorporated side information as hyperpriors to guide the entropy model for more precise modeling. Many subsequent CNN-based works [34], [35], [36], [37] have adopted this VAE framework. Furthermore, significant progress has been made by introducing Transformer architecture into image compression. STF [27] demonstrated the inherent limitations of CNNs in low-level detail restoration and proposed a window-based Transformer. Qian et al. [30] integrated ViT [38] to improve the entropy model's grasp of global context information, whereas Koyuncu et al. [39] used a sliding window technique to simplify the ViT's complexity in the entropy models. TCM [40], on the other hand, has introduced a hybrid approach combining Transformer and CNN in a dual-branch image compression algorithm. FTIC [31] proposed Frequency-Aware Transformer block which performs multiscale directional analysis and frequency

modulation to better capture anisotropic features for learned image compression.

Inspired by the learning-based compression methods commonly used for RGB images, some scholars have begun to integrate learning-based models with the prior knowledge of HSI. He et al. [41] fully leveraged the abundant spectral information and appropriate spectral physical priors by introducing the subspace distillation prior. This innovative prior effectively collaborates with existing learning methods, greatly enhancing the exploration of the spectral low-rank property. References [42], [43], and [44] addressed several challenges in HSI compression, such as non-Gaussian characteristics, compression artifacts, and structural information recovery through techniques like student distribution priors, generative adversarial networks, and side information guidance.

Despite the impressive performance achieved by existing learning-based compression methods, they often overlook the high similarity and correlation in the spectral dimension of HSIs. In this paper, we design a spectral-based self-attention mechanism to capture spectral correlations in HSI and combining it with CNN to extract local spatial details, which better extracts the spectral-spatial features of HSIs compared to the existing methods.

B. Autoregressive Context Models

Drawn from the notion of context found in traditional codecs, Minnen et al. [21] combined autoregressive context with the hyperprior information of images to refine the entropy model, establishing a powerful VAE-based image compression framework with context model. Building upon this, some stronger context models [22], [23], [24], [25] have been subsequently proposed, achieving state-of-the-art RD performance.

However, the serial processing brought by the autoregressive context makes decoding very time-consuming. Therefore, some follow-up works have attempted to achieve a better balance between RD performance and computational efficiency. Minnen and Singh [45] divided latent features into multiple groups along the channel dimension and directly applied context models between channel groups to minimize serial processing. He et al. [46] proposed a checkerboard context model by exploiting spatial correlations in the image, dividing pixels into anchors and non-anchors, where anchors are decoded without relying on context information. Fu et al. [47] proposed a selective kernel residual module and a channel-wise causal context model that split the latent representation into two channel-wise parts and estimated their distributions using gaussian mixture models. Inspired by the observed energy compaction properties in learned image compression, ELIC [48] pioneered a novel uneven channel-conditional adaptive coding strategy, and further incorporated other spatial context model to improve autoregressive efficiency. MLIC [49] built upon the channel-wise context model and accounted the spatial correlations in the RGB images, thus proposed the novel multi-reference entropy model to enhance entropy modeling. Fu et al. [50] made a significant advancement in the checkerboard context model by employing two separate networks to estimate the probability distribution parameters and integrating a more sophisticated probability distribution model

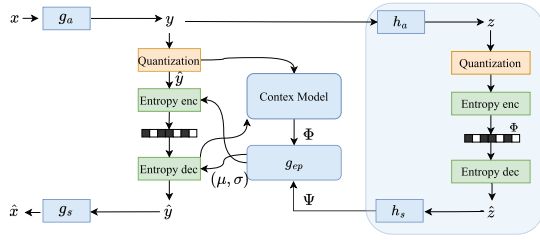


Fig. 1. The architecture of the original context model.

for anchors. And they innovatively introduced deformable convolution and knowledge distillation into image compression to balance the trade-off between performance and efficiency.

Due to the rich spectral information of HSIs further increasing the decoding time of context models, we introduce a more efficient group-wise approach that significantly enhances the parallel processing capability of the context model while ensuring RD performance, achieving a better trade-off between compression performance and speed.

III. METHODOLOGY

A. Formulation

Our work is built upon the context model [21] that is based on VAE, as shown in Fig. 1. We will provide a brief overview of the process to facilitate better understanding.

Given an input hyperspectral image $x \in \mathbb{R}^{h \times w \times c}$, the encoder g_a is first utilized to extract features, resulting in the latent representation y . After quantization by Q , y is compressed into $\hat{y}$ and then decoded by the decoder g_s to obtain the reconstructed image $\hat{x}$. The entire process is formulated as

$$\begin{aligned} y &= g_a(x; \theta_{g_a}), \\ \hat{y} &= Q(y), \\ \hat{x} &= g_s(\hat{y}; \theta_{g_s}), \end{aligned} \quad (1)$$

in which θ_{g_a} and θ_{g_s} represent the trainable parameters of the corresponding modules.

For the quantized $\hat{y}$, it is modeled as a Gaussian distribution by the entropy model and encoded into a bitstream to complete compression. The parameters (μ, σ) of the entropy model are generated by the entropy parameter module g_{ep} parameterized by θ_{ep} based on the hyperprior parameter Ψ and context parameter Φ , where Ψ is obtained through the hyperprior codec from y and Φ is obtained from the autoregressive context, formulated as

$$\mu, \sigma = g_{ep}(\Psi, \Phi; \theta_{ep}). \quad (2)$$

The objective of the compression task is to minimize the distortion D between $\hat{x}$ and x , while keeping the bitrate R as small as possible. So the loss function can be represented as

$$\begin{aligned} \mathcal{L} &= R(\hat{y}) + R(\hat{z}) + \lambda \cdot D \\ &= -\mathbb{E}_{x \sim p_x} [\log_2 p_{\hat{y}}(\hat{y}) + \log_2 p_{\hat{z}}(\hat{z})] + \lambda \cdot \mathbb{E}_{x \sim p_x} [d(x, \hat{x})]. \end{aligned} \quad (3)$$

where λ controls the trade-off between rate R and distortion D . The bitrate R is estimated as the negative log-likelihood of the probability distribution p , and $\hat{z}$ is the quantized hyperprior latent variable obtained from the hyperprior encoder. The

entropy model estimates the probability distributions $p_{\hat{y}}(\hat{y})$ and $p_{\hat{z}}(\hat{z})$ for bitstream encoding.

B. Overall Architecture

The aforementioned architecture achieves more precise entropy modeling by combining autoregressive context with hyperprior information. However, its CNN-based codec structures cannot fully capture the spectral dependencies of HSIs, thereby limiting their performance in HSI compression task. In addition, the autoregressive context inevitably introduces serial processing, which significantly reduces coding efficiency for HSIs.

Therefore, we propose an architecture tailored for HSI compression, as illustrated in Fig. 2. To take advantages of the spectral characteristics of HSIs, we first propose the SSCTG to compose the codecs g_a , g_s , h_a , h_s , which uses Transformer-CNN hybrid structure to better extract the spectral-spatial coupling features of HSIs. Each encoder or decoder consists of N SSCTGs. The HSI x is first compressed into a more compact latent representation y through g_a , and further extracted into the edge information z by the hyperprior encoder h_a . After obtaining $\hat{z}$ through quantization and entropy model, two hyperprior decoders $h_{s,\sigma}$ and $h_{s,\mu}$ are used to generate hyperprior parameters Ψ_σ and Ψ_μ for subsequent parameter generation in g_{ep} . Meanwhile, the quantized $\hat{y}$ undergoes compression through the autoregressive entropy model, and is then reconstructed into image $\hat{x}$ through the decoder g_s .

Considering the inefficiency caused by the autoregressive context in the above process, we further propose the GCM to parallelize autoregression. Specially, we divide the quantized latent representation $\hat{y}$ into K groups along the channel dimension and perform autoregression in parallel across each group to obtain the context parameters $\Phi_i^{(k)}$. Together with Ψ_σ and Ψ_μ , they jointly predict the parameters σ and μ of the entropy model. For a latent representation $\hat{y} \in \mathbb{R}^{h \times w \times c}$, our proposed GCM adopts a group-wise and channel-wise approach, requiring only $\frac{c}{K}$ steps to complete the autoregressive process of the entropy model, instead of autoregressing pixel by pixel as in the original context model.

In addition, to effectively preserve the spectral information, we introduce Spectral Angle Mapping (SAM) to measure spectral distortion as

$$L_{\text{SAM}}(x, \hat{x}) = \frac{1}{h \times w} \times \sum_{j=1}^{h \times w} \cos^{-1} \left(\frac{x_j \otimes \hat{x}_j}{\|x_j\| \|\hat{x}_j\|} \right), \quad (4)$$

in which $\otimes$ denotes inner product. $\|\cdot\|$ denotes the L2 norm, x_j and $\hat{x}_j$ represent the j -th pixel of the original and reconstructed image, respectively.

Thus, with L_{SAM} to supervise the network in training process, the proposed loss function can be formulated as

$$\begin{aligned} \mathcal{L} &= -\mathbb{E}_{x \sim p_x} [\log_2 p_{\hat{y}}(\hat{y}) + \log_2 p_{\hat{z}}(\hat{z})] \\ &\quad + \lambda \cdot (L_{\text{MSE}}(x, \hat{x}) + \alpha \cdot L_{\text{SAM}}(x, \hat{x})). \end{aligned} \quad (5)$$

Similar to previous works [13], [21], [33], we add average noise into y during the training process to replace quantization so that all modules can participate in end-to-end training.

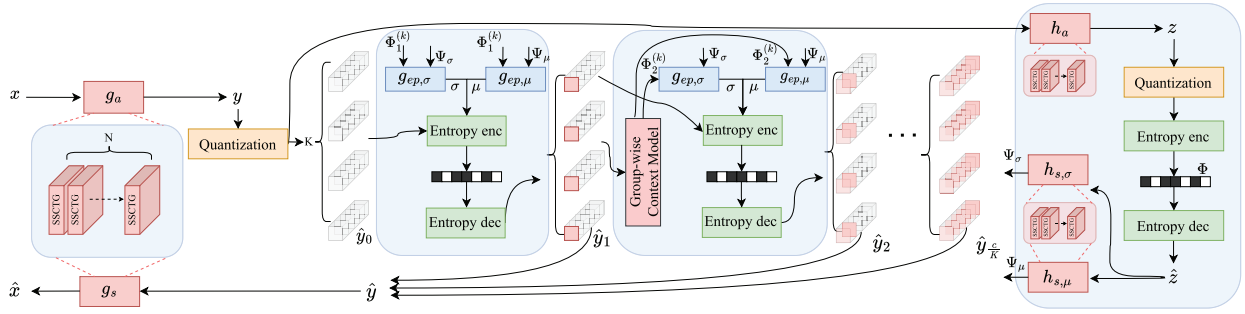


Fig. 2. The overall architecture of the proposed method. We propose SSCTG to construct the codecs and GCM to further improve the efficiency of entropy codecs.

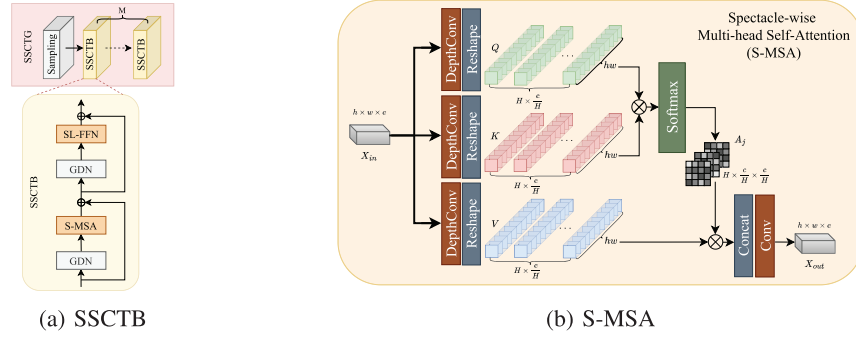


Fig. 3. (a) The specific structure of SSCTB used to compose SSCTG. (b) S-MSA used to capture spectral correlations.

In the following two sections, we will provide a detailed introduction to the proposed Spectral-Spatial Coupling Transformer Group and Group-wise Context Model.

C. Spectral-Spatial Coupling Transformer Group (SSCTG)

Most existing learning-based compression methods have primarily focused on spatial correlations, while largely neglecting the high redundancy and correlation present in the spectral dimension. Due to the inherent limitations of convolution in capturing spectral similarity [38], [51], [52], we aim to explore the Transformer architecture to better capture the spectral correlation of HSIs. However, directly applying the original Transformer faces two challenges. On the one hand, Transformer exhibits strong modeling capabilities for long-range dependencies in spatial dimensions, whereas HSIs are typically sparse in spatial dimensions but exhibit strong dependencies in the spectral dimension. On the other hand, HSIs often exhibit strong correlations between adjacent or nearby pixels [27], [46], whereas Transformer is less adept at modeling such local spatial information.

Therefore, we propose the SSCTG to construct the codecs. Each SSCTG consists of a downsampling or upsampling module and M Spectral-Spatial Coupling Transformer Blocks (SSCTB), as shown in Fig. 3(a), which work together to extract spectral-spatial coupled features from HSIs. Unlike existing deep learning-based HSI compression models those rely solely on convolutional networks, in SSCTB, we introduce Spectral-wise Multi-head Self-Attention (S-MSA) to effectively capture the spectral features of HSIs. Additionally, we propose the CNN-based Spatial Local FeedForward Network (SL-FFN) to model local information in HSIs, capturing spatial local

dependencies. Moreover, we integrate Generalized Divisive Normalization [53] (GDN) to the network to normalize the features, as GDN applies a nonlinear transformation to adaptively adjust to the data distribution, effectively reducing noise introduction and better preserving high-frequency details compared to Batch Normalization or Layer Normalization. In particular, for the SSCTB in the hyperprior codec h_a and h_s , we have removed the residual connections to simplify the network and accelerate training.

1) *Spectral-Wise Multi-Head Self Attention (S-MSA)*: Due to the strong spectral correlation in HSIs, we propose S-MSA to capture spectral dependencies. S-MSA treats the spectral vector of each pixel as the input to the module and computes self-attention along the spectral dimension, as shown in Fig. 3(b). The input features $X_{in} \in \mathbb{R}^{h \times w \times c}$ are transformed by depth-wise convolutions to generate query vectors $Q \in \mathbb{R}^{hw \times c}$, key vectors $K \in \mathbb{R}^{hw \times c}$, and value vectors $V \in \mathbb{R}^{hw \times c}$ as

$$\begin{cases} Q = W^Q X_{in}, \\ K = W^K X_{in}, \\ V = W^V X_{in}. \end{cases} \quad (6)$$

W^Q, W^K, W^V are learnable parameters. Then, the vectors Q, K , and V are split into H groups along the spectral dimension as $Q = [Q_1, \dots, Q_H], K = [K_1, \dots, K_H], V = [V_1, \dots, V_H]$, where each group has a dimension of $d_h = \frac{c}{H}$, and H denotes the number of heads. Next, the self-attention along the spectral dimension is computed for each head as follows.

$$A_j = \text{softmax}(\sigma_j K_j^T Q_j), \quad (7)$$

$$\text{head}_j = V_j A_j. \quad (8)$$

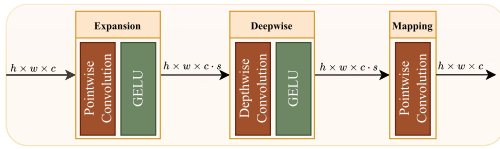


Fig. 4. The architecture of SL-FFN.

(Q_j, V_j) represent the vectors of the j -th group, and K_j^T denotes the transpose of K_j . Since the spectrum varies significantly with the wavelength, a learnable parameter $\sigma_j \in \mathbb{R}^1$ is utilized to weight $K_j^T Q_j$ adaptively, to learn the attention map A_j for each head. Finally, the outputs of H heads are concatenated along the spectral dimension and projected to obtain the output $X_{out} \in \mathbb{R}^{h \times w \times c}$ as

$$X_{out} = (\text{Concat}_{j=1}^N(\text{head}_j))W, \quad (9)$$

in which $W \in \mathbb{R}^{c \times c}$ is a learnable parameter.

Based on the spectral dimension, the S-MSA computes self-attention along the spectral dimension of latent features, effectively capturing dependencies among spectra. Furthermore, grouping spectra into multiple heads can also reduce the computational complexity of self-attention.

2) Spatial Local FeedForward Network (SL-FFN):

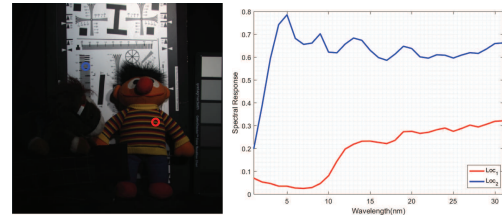
Although S-MSA captures more compact spectral representations by modeling correlations among spectra, the linear layer-based feed-forward network in the original Transformer lacks the ability to capture spatial local details effectively. Due to the strong correlations in adjacent spatial locations of HSIs, we propose SL-FFN to further enhance the local spatial information modeling capability of SSCTB.

As shown in Fig. 4, SL-FFN consists of an expansion layer, a depthwise layer and a mapping layer. To minimize computational complexity, the depthwise layer utilizes 3×3 depthwise convolution to extract spatial local features from HSIs. However, applying convolutional layers to low-dimensional tensors may not extract sufficient information. Hence, an expansion layer is added before the depthwise layer to map the input feature vector $f_{in} \in \mathbb{R}^{h \times w \times c}$ from a low-dimensional space to a high-dimensional $\mathbb{R}^{h \times w \times c \cdot s}$ space using 1×1 pointwise convolution. Finally, the mapping layer utilizes 1×1 pointwise convolution to map the high-dimensional tensor back to a low-dimensional space.

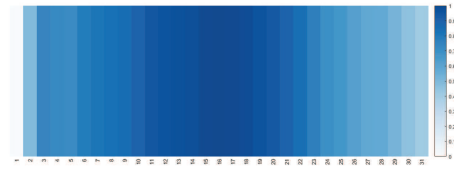
D. Group-Wise Context Model (GCM)

Autoregression defines a joint distribution based on the condition of each feature, assuming that the value of each feature depends on the value of previous features. Reference [21] combines autoregression with VAE, in which the i -th pixel $\hat{y}_i$ is predicted based on its preceding $(i - 1)$ -th elements.

However, as mentioned earlier, constrained by the strict serial processing requirement of autoregression, image must be decoded in the order of $\hat{y}_1, \dots, \hat{y}_{hwc}$, leading to a time complexity linearly related to the image size. As illustrated in Fig. 6(a), the workflow of the original context model is constrained by autoregression, allowing only one pixel to be decoded in a single step, resulting in inefficiency of the process. This issue will be further exacerbated in HSI



(a) Image shown in RGB (b) Spectral curves



(c) Spectral correlation

Fig. 5. Spectral curves and spectral correlation illustration for the HSI image chart_and_stuffed_toy in the CAVE dataset.

compression because HSIs have more spectral channels than conventional RGB images.

Therefore, considering the correlation between adjacent spectra in HSIs, we propose GCM to enhance the parallel processing capability of decoding. We partition the HSIs along the channel dimension into groups and apply the autoregression within each channel group to perform parallel decoding across spectral groups.

1) *Spectral Correlation Analysis*: Due to the imaging process of HSIs and the small differences in wavelength between adjacent spectra, the spectral response of a pixel typically exhibits a continuous curve, as shown in Fig. 5(b). Furthermore, the spectral correlation between the i -th and $(i + 1)$ -th spectral bands of the HSI can be defined as $r_{(i,i+1)} =$

$$\frac{\sum_{m=1}^h \sum_{n=1}^w [f_i(m, n) - \bar{f}_i][f_{i+1}(m, n) - \bar{f}_{i+1}]}{\sqrt{\left(\sum_{m=1}^h \sum_{n=1}^w [f_i(m, n) - \bar{f}_i]^2 \right) \left(\sum_{m=1}^h \sum_{n=1}^w [f_{i+1}(m, n) - \bar{f}_{i+1}]^2 \right)}}. \quad (10)$$

where h and w represent the height and width of the HSI. $f_i(m, n)$ represents the pixel values at position (m, n) for the i -th spectral band, $\bar{f}_i$ denotes the mean value of all pixels in the i -th spectral band.

As an example, Fig. 5(c) illustrates the correlation between the 16-th spectral band and other bands within a representative image (e.g., the “chart_and_stuffed_toy” image from the HSI dataset CAVE [54]). It can be seen that spectral bands closer in distance exhibit darker colors, indicating stronger correlation. While those farther apart show lighter colors, signifying weaker correlation. Therefore, modeling based on channel-wise autoregressive entropy model [45] may lead to inaccuracy due to the weaker correlation between distant channel groups, thereby influencing compression performance. Hence, we propose modeling the context for channels closer in distance within each channel group, thereby enhancing parallel processing capability while ensuring compression performance.

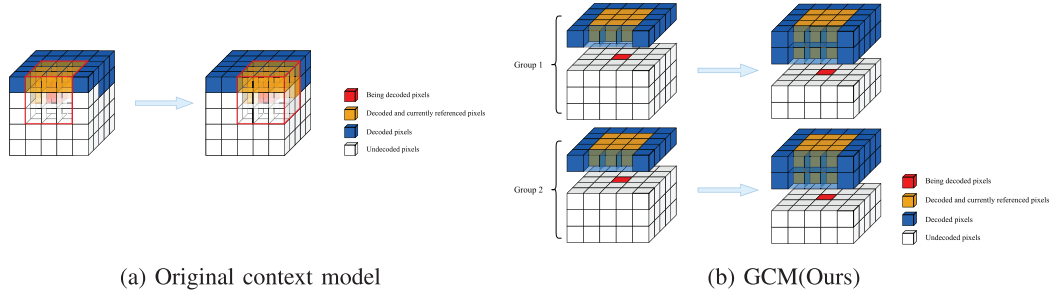


Fig. 6. The workflow of the original context model and our proposed group-wise context model.

2) *Group-Wise Context Model (GCM)*: Due to the stronger correlation among closer channels, we propose GCM to model the contextual dependencies among nearby channels within each group. The workflow of GCM is illustrated in Fig. 6(b), where we first partition the latent feature $\hat{y} \in \mathbb{R}^{h \times w \times c}$ along the channel dimension into K groups $\hat{y} = [\hat{y}^{(1)}, \hat{y}^{(2)}, \dots, \hat{y}^{(K)}]$, where each tensor slice has dimensions $\hat{y}_i^{(k)} \in \mathbb{R}^{h \times w \times \frac{c}{K}}$ ($k = 1, 2, \dots, K$). Then, autoregressive prediction is employed within each group. For the k -th sliced group, GCM generates contextual parameters $\Phi_i^{(k)}$ based on the $(i - 1)$ decoded channels within the group, which are used to subsequently generate entropy parameters for encoding and decoding the i -th channel. Therefore, the prediction $\hat{y}_i^{(k)}$ for the i -th channel depends on its preceding $(i - 1)$ channels within the group, described by the following equation.

$$\begin{aligned} \hat{y}_i^{(k)} &\leftarrow \{\hat{y}_1^{(k)}, \hat{y}_2^{(k)}, \dots, \hat{y}_{i-1}^{(k)}\}, \\ \Phi_i^{(k)} &= \begin{cases} 0, & i = 1 \\ GCM(\hat{y}_{<i}^{(k)}; \theta_{cm}^{(k)}), & i \neq 1. \end{cases} \end{aligned} \quad (11)$$

θ_{cm} represents the trainable parameters of the GCM. It is worth noting that when predicting the first channel of each group, since there are no previously decoded channels, the context parameters will be set to 0.

After obtaining the context parameters $\Phi_i^{(k)}$ by GCM, they collectively predict the parameters σ and μ of the entropy model along with hyperprior parameters Ψ_σ and Ψ_μ generated by hyperprior decoders $h_{s,\sigma}$ and $h_{s,\mu}$, as

$$\begin{aligned} \sigma_i^{(k)} &= g_{ep,\sigma}(\Psi_\sigma, \Phi_i^{(k)}; \theta_{ep,\sigma}^{(k)}), \\ \mu_i^{(k)} &= g_{ep,\mu}(\Psi_\mu, \Phi_i^{(k)}; \theta_{ep,\mu}^{(k)}). \end{aligned} \quad (12)$$

$g_{ep,\cdot}$ denotes the module used for generating the entropy model parameters, and $\theta_{ep,\cdot}^{(k)}$ represents its network parameters.

In general, the parallel processing capability of GCM increases with the number of groups K . In a GCM with K groups, each group can be decoded in parallel, and within each group, decoding is performed channel-wise. Therefore, GCM can process $h \times w \times K$ pixel values in parallel. This significantly enhances the parallel processing capability of the model compared to the original autoregressive context model, which can only process one pixel at a time. However, as the number of groups increases, the number of vectors within each group decreases, which may lead to a decrease in compression performance. Therefore, when selecting the number of groups K , a balance between the performance and efficiency of the

model should be considered. Ablation experiments on the number of groups K can be found in Table II.

IV. EXPERIMENTS

A. Experimental Setup

Datasets: We selected two publicly available HSI datasets, CAVE [54] and NTIRE [56], to validate the effectiveness of the proposed method. Both datasets consist of 31 spectral bands covering the wavelength range from 400nm to 700nm. The CAVE dataset comprises indoor scene HSIs, including 32 images with spatial dimensions of 512×512 . The NTIRE dataset contains 950 HSIs, each with spatial dimensions of 512×482 .

Evaluation Metrics: To measure image distortion, we adopted Peak Signal-to-Noise Ratio (PSNR), Multiscale Structural Similarity Index Metric (MS-SSIM), and SAM to comprehensively evaluate the reconstruction quality. In addition, we used Bjontegaard Delta rate (BD-rate) to measure the relative performance of different methods.

1) *PSNR*: PSNR compares the reconstructed and original images at the pixel level, with a higher value indicating better fidelity.

2) *MS-SSIM*: MS-SSIM measures the structural similarity between two images, with values ranging from 0 to 1, where higher values indicate greater similarity. For easier comparison, following previous studies [13], we apply a logarithmic transformation to convert MS-SSIM into a decibel (dB) scale, which emphasizes differences when values are close to 1.

3) *SAM*: SAM assesses spectral accuracy by computing the mean spectral angle between corresponding positions in the reconstructed and ground-truth images, with a smaller value indicating higher fidelity.

4) *BD-Rate*: BD-rate represents the relative difference in bit-rate required by two compression methods to achieve the same image quality, typically measured by PSNR. A negative BD-rate indicates better relative compression performance.

Comparison Methods: In Subsection IV-B, we compared our proposed method with traditional HSI compression approach KLT+JPEG2000 [55] and the state-of-the-art deep learning-based compression methods [21], [22], [27], [33], [40], [48], [49], where Context [21] was the baseline model. Among them, Hyperprior [33], Context [21], Chengatt [22], ELIC [48] and MLIC [49] employed CNN-based codecs. While STF [27] utilized a window-based Transformer module to extract image features and TCM [40] proposed a

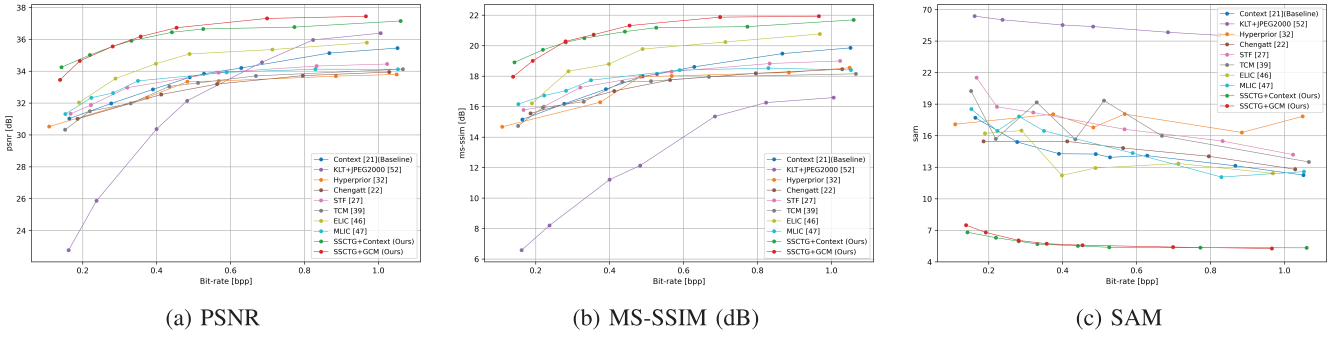


Fig. 7. RD curves in terms of PSNR, MS-SSIM, and SAM on the CAVE dataset.

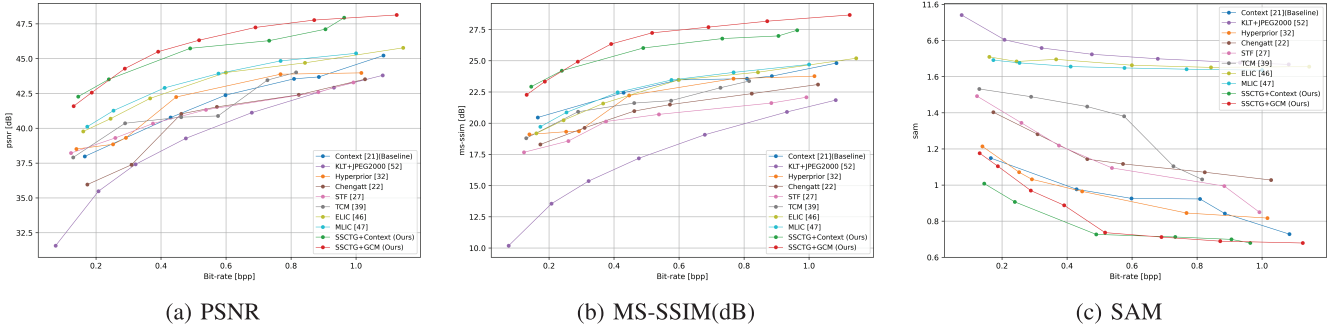


Fig. 8. RD curves in terms of PSNR, MS-SSIM, and SAM on the NTIRE dataset.

dual-branch image compression algorithm using both Transformer and CNN.

In Subsection IV-C, we further compared the proposed GCM with different context models [21], [45], [46], [48], [49] when utilizing the same proposed SSCTG network structure.

In our method, we use a grouping number of $K = 8$ to achieve better RD performance. For [45], we adopt a 16-slices setting instead of the original 10-slices configuration in order to achieve better performance on HSIs, while other comparison methods follow the default optimal settings reported in their respective papers. We conducted a comprehensive comparison of these context models in terms of compression quality and encoding/decoding efficiency.

In Subsection IV-D, we fixed the context modeling using the proposed GCM and conducted comparative experiments between SSCTG and several other network architectures, including STF [27], TCM [40], and FTIC [31].

Implementation Details: We trained the learning-based methods based on the Pytorch framework and obtained different compression rates by adjusting the λ in the loss function. We applied the same data augmentation techniques to all methods, including random cropping, random flipping, and random rotation. We utilized Adam [57] to optimize the networks with a batch size of 16. The initial learning rate was set to 1×10^{-4} and decayed at the 200th and 500th epochs.

B. Comparison With the Sota Methods

1) RD Performance: Fig. 7 illustrates the compression results of all methods on the CAVE dataset at different compression rates, where SSCTG+Context represents using our proposed SSCTG to build the codecs based on the

baseline Context [21] model. And SSCTG+GCM represents our proposed overall method which further integrates the proposed GCM. Across all bit-rates, SSCTG+Context achieves significantly higher PSNR, MS-SSIM and SAM than other comparison methods, demonstrating superior RD performance on HSI compression. In addition, SSCTG+GCM which integrates GCM to refine the autoregressive process can further improve the overall PSNR and MS-SSIM of the reconstructed images, particularly noticeable at high bit-rates.

Fig. 8 illustrates the RD curves of all methods on the NTIRE datasets. Our SSCTG+Context and SSCTG+GCM still significantly outperform other methods, yielding similar conclusions to those on the CAVE dataset. And SSCTG+GCM achieves even higher compression quality from medium to high bit-rates compared to SSCTG+Context.

2) Visual Comparison: Fig. 9 shows the reconstruction results of the chart_and_stuffed_toy image from the CAVE dataset at similar bit-rates. The first row displays the reconstructed images for all methods and we present the images in RGB format for better visual comparison. The second row represents the residual maps between the reconstructed image and the ground-truth image. Colors closer to dark blue indicate better reconstruction performance, while colors closer to red indicate poorer reconstruction performance.

The reconstruction quality of KLT+JPEG2000 is relatively uniform across the entire image. CNN-based methods including Hyperprior, Context, Chengatt, ELIC, MLIC exhibit artifacts due to inherent limitations of convolution. Among them, ELIC shows noticeable stripe artifacts in the reconstructed image, as evidenced by the magnified regions. This indicates that although ELIC proposes an alternative grouped

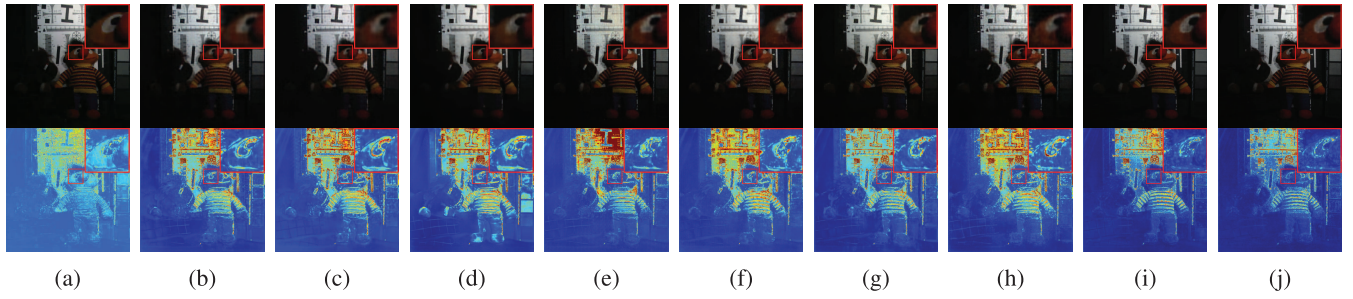


Fig. 9. The visual quality comparison among different methods on the CAVE dataset. (a) KLT+JPEG2000 [55]. (b) Hyperprior [33]. (c) Context [21] (Baseline). (d) Chengatt [22]. (e) STF [27]. (f) TCM [40]. (g) ELIC [48]. (h) MLIC [49]. (i) SSCTG+Context (Ours). (j) SSCTG+GCM (Ours).

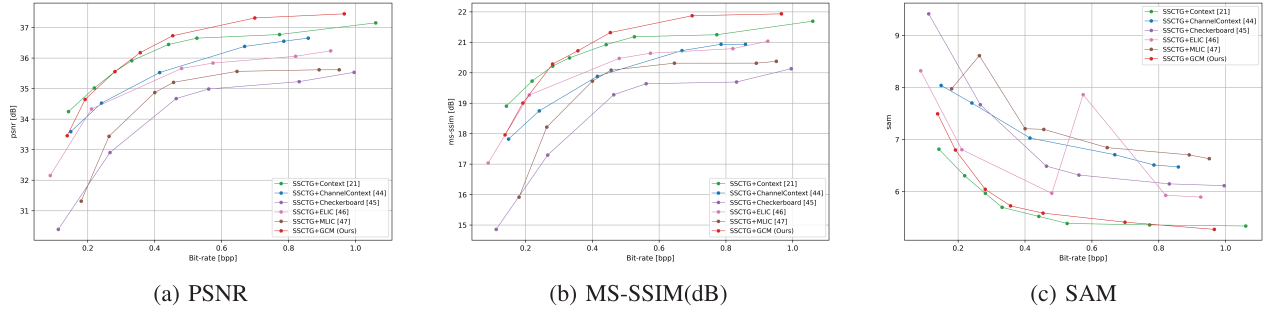


Fig. 10. Performance comparison of GCM and other context models in combination with SSCTG on the CAVE dataset.

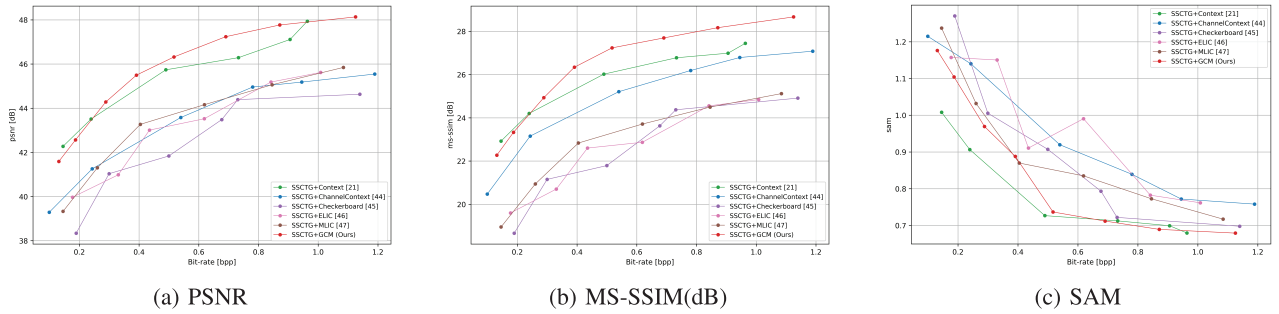


Fig. 11. Performance comparison of GCM and other context models in combination with SSCTG on the NTIRE dataset.

parallel strategy based on observations of energy compaction in process of RGB image compression, it fails to account for the strong correlation between adjacent spectral channels in HSI, leading to less accurate context modeling. Our method achieves better visual quality compared to the aforementioned methods and also outperforms STF and TCM which use Transformer or Transformer-CNN hybrid architecture.

C. Performance and Coding Efficiency of Context Models

1) Performance of Different Context Models Compared With GCM: To better demonstrate the advantages of the proposed GCM over other context models in HSI compression, we conducted comparative experiments by combining the proposed GCM and several other context models with SSCTG. As shown in Fig. 10 and Fig. 11, our proposed GCM achieves the best performance in terms of PSNR, MS-SSIM and SAM across all evaluated context models. The experimental results demonstrate that the proposed GCM enables the context model to perform more accurate autoregression based on spectrally

dependent channels, thereby improving HSI compression performance. GCM groups nearby channels for autoregressive modeling, improving both parallelism and performance. In contrast, while some other improved context models (e.g., ChannelContext, ELIC, etc.) also use parallel strategies, their performance on HSI is worse than GCM, as they neglect the spectral dependencies inherent in HSIs.

2) Coding Efficiency of Different Context Models:

Although autoregressive context models significantly improve the RD performance, the sequential prediction of latent representations leads to lower efficiency. To verify the improvement in coding efficiency brought by GCM while maintaining performance, we conducted a comprehensive comparison of BD-rate and inference latency on a single RTX 3090 GPU by combining different context models with SSCTG.

Table. I illustrates the results, where TEnc and TDec represent the total encoding and decoding times, and YEnc and YDec denote the time for entropy encoding and decoding of the latent feature $\hat{y}$. Our proposed GCM significantly improves the efficiency of entropy encoding and decoding for

TABLE I

COMPARISON OF ENCODING/DECODING TIME ON SINGLE RTX 3090 GPU FOR DIFFERENT CONTEXT MODELS IN COMBINATION WITH SSCTG

Method (SSCTG+)	CAVE					NTIRE				
	BD-rate (%)	TEnc	TDec	YEnc	YDec	BD-rate (%)	TEnc	TDec	YEnc	YDec
Context [21]	0.00	3.503	6.125	3.378	6.074	0.00	3.481	6.164	3.427	6.129
ChannelContext [45]	47.12	0.245	0.186	0.119	0.133	116.82	0.161	0.179	0.102	0.142
Checkerboard [46]	179.85	0.160	0.081	0.031	0.032	202.36	0.084	0.067	0.025	0.033
ELIC [48]	54.77	0.334	0.132	0.082	0.084	124.44	0.125	0.127	0.061	0.085
MLIC [49]	99.47	0.413	0.406	0.090	0.158	102.36	0.146	0.393	0.067	0.159
GCM	-7.36	0.230	0.186	0.107	0.134	-11.04	0.153	0.173	0.095	0.136

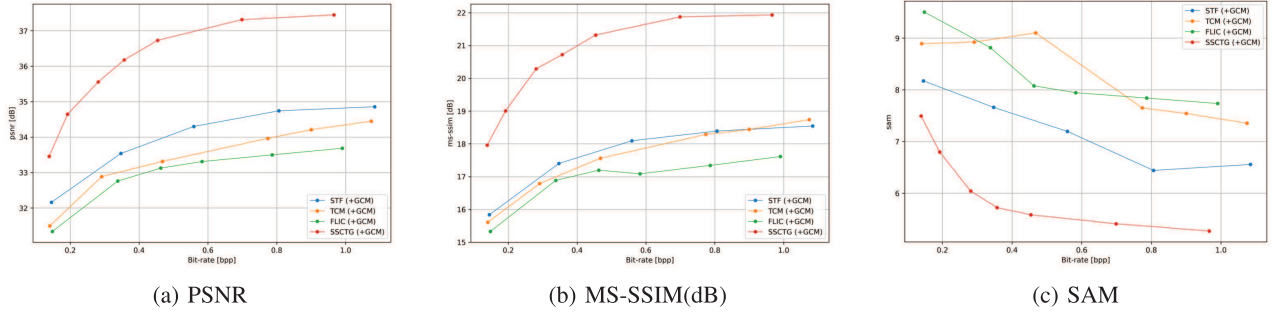


Fig. 12. Performance comparison of SSCTG and other network architectures on the CAVE dataset.

the latent feature $\hat{y}$. Taking SSCTG+Context as the baseline method, SSCTG+GCM obtains the best BD-rate over all other context models. Compared to the baseline context model, it not only improves the RD performance but also reduces the entropy encoding time from 3.378s to 0.107s and the entropy decoding time from 6.074s to 0.134s on the CAVE dataset, achieving a significant efficiency improvement. This is because our proposed GCM divides the latent features into K groups along the channel dimension and all groups are processed in parallel. Furthermore, the intra-group sequential process is conducted along the channel dimension rather than pixel-by-pixel as in Context. As a result, the proposed GCM is able to decode $H \times W \times K$ latent values simultaneously at each step, significantly enhancing decoding efficiency on GPU.

Although some context models (e.g., Checkerboard, etc.) have faster encoding and decoding times, they come at the cost of sacrificing performance. In contrast, GCM can significantly improve efficiency while maintaining compression performance.

D. Performance of SSCTG in Comparison With Other Network Architectures

To further demonstrate the superiority of SSCTG on HSIs, we decoupled SSCTG from the autoregressive entropy model and conducted independent comparative experiments on it. We fixed the context modeling using the proposed GCM and conducted comparative experiments with several other network architectures that incorporate Transformer modules, including STF, TCM, and FTIC.

Although all three comparison methods incorporate Transformers into their architectures, they are not well-suited to

simultaneously capture the spectral and spatial information in HSIs. STF and TCM employ window-based self-attention mechanisms to better capture spatial features, while FLIC designs a frequency-aware Transformer to extract anisotropic features. In contrast, our method leverages a spectral-wise self-attention mechanism to better model the spectral characteristics of HSIs, and replaces linear layers with local CNNs in Transformer to enhance the extraction of spatial local information.

As shown in Fig. 12 and Fig. 13, under the fixed GCM, our proposed SSCTG consistently achieves better RD performance compared with other methods across two HSI datasets. This further demonstrates the superior capability of SSCTG in capturing spatial-spectral redundancy and provides decoupled evidence supporting the effectiveness of its architectural design.

E. Ablation Study

To further validate the effectiveness of each module of the proposed method, we conducted corresponding ablation experiments on the CAVE dataset.

1) *Effectiveness of SSCTG and GCM*: To verify the effectiveness of the main modules proposed in this paper, we conducted ablation experiments on the SSCTG and GCM, as shown in Fig. 14. By incorporating SSCTG and GCM into the baseline model, we observed significant improvements in RD performance, with adding both SSCTG and GCM showing the best overall performance, particularly at higher bpp. The experimental results validate the effectiveness and adaptability of SSCTG and GCM for HSI compression, demonstrating

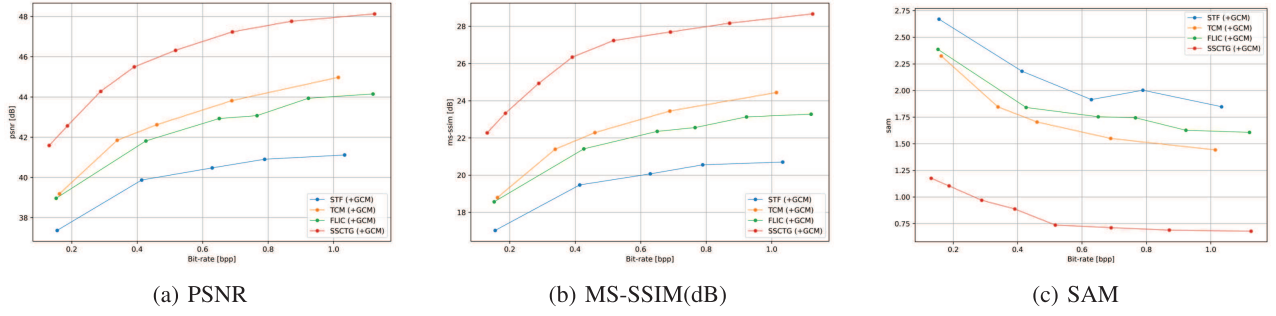


Fig. 13. Performance comparison of SSCTG and other network architectures on the NTIRE dataset.

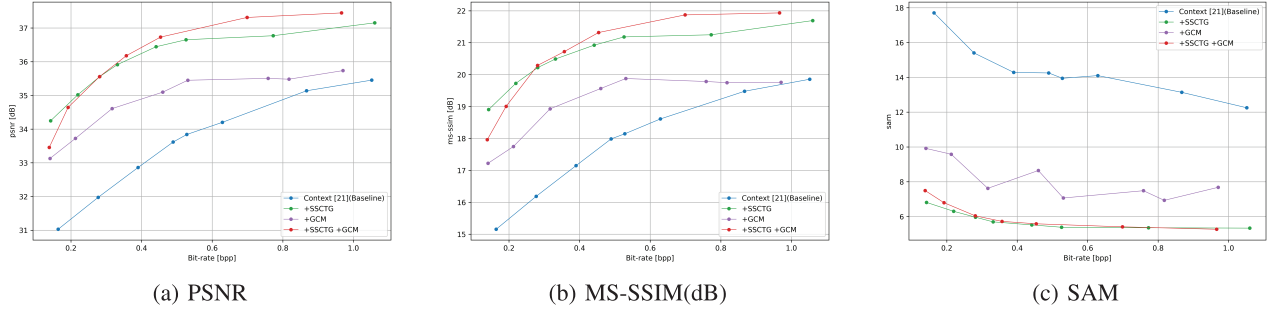


Fig. 14. Ablation study on proposed SSCTG and GCM.

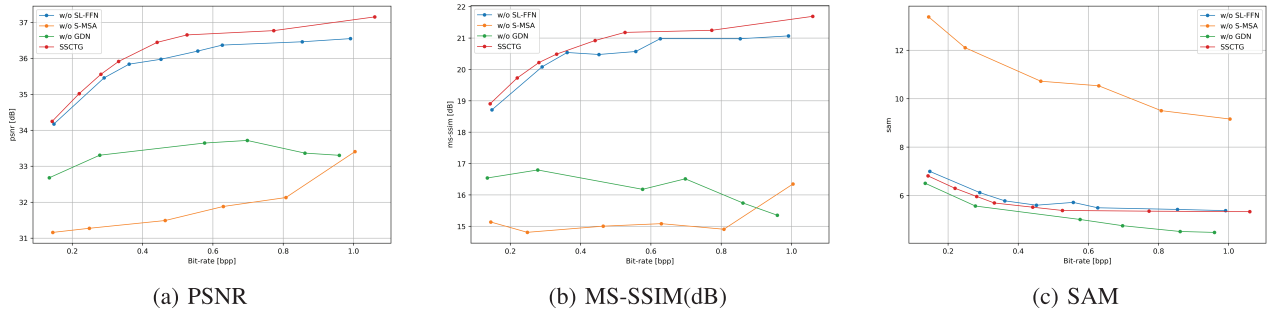


Fig. 15. The impact of each module in SSCTG on the compression performance.

their ability to effectively capture and utilize the unique characteristics of HSIs.

2) *Effectiveness of Each Module in SSCTG*: We conducted ablation experiments on each module of SSCTG, as shown in Fig. 15. Specifically, w/o S-MSA refers to replacing the proposed S-MSA with spatial dimension-based multi-head self-attention of the original Transformer. W/o SL-FFN refers to replacing the proposed SL-FFN with linear layer-based feed-forward network of the original Transformer, and w/o GDN adopts Layer Normalization instead of GDN used in this paper. According to the experimental results, it is concluded that removing both S-MSA and SL-FFN will lead to a decrease in the compression performance of the model. While removing GDN, although it enhances the spectral reconstruction ability to some extent, it results in a decrease in the overall compression performance on spatial domain, with a noticeable performance degradation in both PSNR and MS-SSIM.

3) *Impact of the Number of Groups in GCM on Performance*: In the proposed GCM, the latent features are divided into K groups along the channel dimension. The parallelism of

the model increases as the number of groups K increases. But at the same time, the number of vectors within each group decreases, which may lead to a reduction on compression performance.

To investigate the impact of the number of groups K on model performance, we conducted ablation experiments with different K , as shown in Fig. 16. When the number of groups is small, the compression performance of the model does not differ significantly. However, as the number of groups increases, such as when $K = 64$, each group only contains several channels, and the available contextual information within each group is limited, leading to a sharp decline in compression performance.

Specially, considering the case of $K = 1$, the GCM essentially corresponds to the case of “no grouping”. In this case, the entire latent representation is modeled autoregressively along the channel dimension on a per-channel basis. Under this setting, our method degenerates to the special case in [45], where each channel is treated as an individual slice. Without grouping, spectral channels that are far apart may introduce

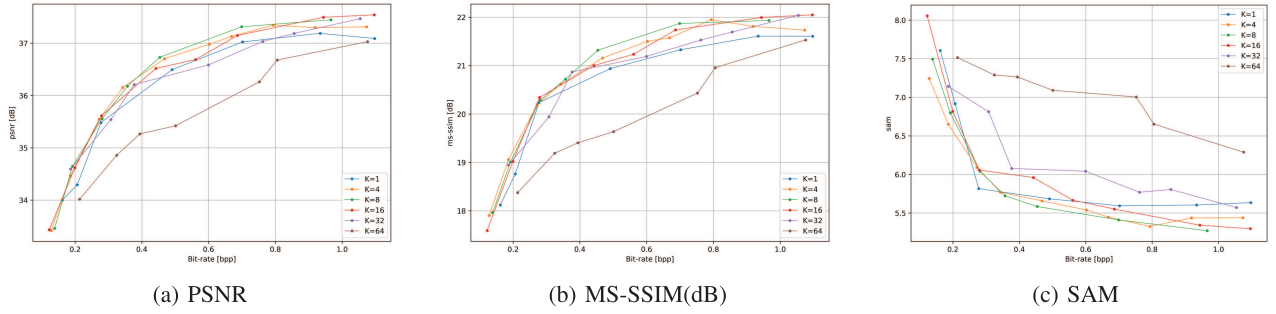


Fig. 16. The impact of the number of groups K in GCM on the compression performance.

TABLE II
IMPACT OF THE NUMBER OF GROUPS K IN GCM
ON ENCODING/DECODING EFFICIENCY

$K=$	BD-rate (%)	Inference Latency (s)			
		TEnc	TDec	YEnc	YDec
1	6.63	0.669	0.908	0.593	0.851
4	-6.83	0.308	0.296	0.181	0.244
8	-7.36	0.230	0.186	0.107	0.135
16	-3.03	0.197	0.128	0.070	0.077
32	2.24	0.175	0.099	0.051	0.048
64	65.37	0.165	0.084	0.043	0.033

noise into the autoregressive context modeling, resulting in less accurate entropy estimation and degraded RD performance. As shown in the experimental results, the RD performance of the ungrouped case ($K = 1$) is inferior to that of the grouped cases with $K = 4, 8, 16$, and 32 . More importantly, the case of $K = 1$ significantly reduces the parallelism of the model, resulting in a substantial drop in efficiency.

Moreover, [47] divides the latent features into two groups for joint modeling, but the method is designed for RGB images. As the number of groups increases, the performance degrades, so only two groups can be used, which limits its parallelism. In contrast, our proposed method achieves both better RD performance and higher parallelism as the group number K increases within a reasonable range. Overall, the model achieves the best compression performance when $K = 8$. Therefore, we set $K = 8$ as the number of groups in GCM for other experiments.

4) *Impact of the Number of Groups in GCM on Encoding/Decoding Efficiency*: To verify the influence of different numbers of groups K in GCM on compression efficiency, we conducted ablation experiments on GPU shown in Table II. It is observed that with the increase in K , the overall time for encoding and decoding decreases, with the reduction mainly attributed to the shorter times for entropy encoding and decoding (YEnc and YDec) of the latent feature $\hat{y}$. Specifically, in the ungrouped case (i.e., $K = 1$), the encoding/decoding time is significantly higher than in the grouped settings. As the number of groups increases in the GCM, the model is able to compute across multiple groups simultaneously, which aligns well with the parallel computing capabilities of GPUs.

However, the excessively large number of groups lead to a reduction in the average number of channels context model

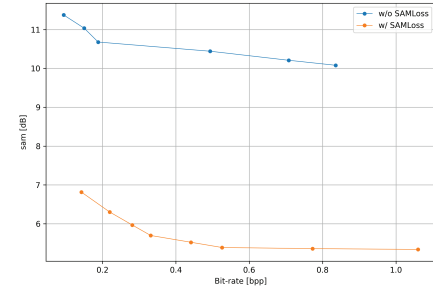


Fig. 17. The ablation study on the proposed method w/and w/o SAMLoss.

utilized for prediction, resulting in a decrease in obtaining contextual information and thus influence the performance. Therefore, when applying GCM for grouping in the context model, an appropriate number of groups K should be chosen to balance the trade-off between performance and encoding/decoding efficiency.

5) *Impact of the SAMLoss*: In our proposed method, we incorporate SAM as part of the image distortion loss D in training process, referred to as SAMLoss. By introducing SAMLoss, we aim to explicitly constrain spectral distortion in addition to spatial distortion, thereby enhancing the network's spectral reconstruction ability. To verify the effectiveness of SAMLoss in spectral reconstruction, we conducted an ablation study with and without SAMLoss, as shown in Fig. 17. Introducing SAMLoss significantly reduces the spectral angle difference between the reconstructed and ground truth images, enhancing spectral reconstruction performance.

V. CONCLUSION

In this paper, we have extensively investigated the application of the Transformer architecture in hyperspectral image (HSI) compression, proposing the Spectral-Spatial Coupling Transformer Group (SSCTG) as a plug-and-play component for constructing the codecs. We proposed to utilize spectral-wise multi-head self-attention and spatial local feed-forward networks to capture the spectral-spatial coupling features in HSI. Additionally, we introduced the Group-wise Context Model (GCM), which significantly enhances the parallel processing capability of the autoregressive entropy model by performing autoregression on the latent features of HSIs grouped along the spectral dimension. Extensive experimental results demonstrate the effectiveness of the proposed approach,

surpassing state-of-the-art RD performance while exhibiting high efficiency. In the future, exploring the relationships between adjacent channel groups in HSIs may further enhance the performance and efficiency of the context model.

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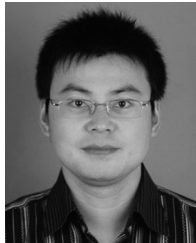


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